

CS 1671 / CS 2071

Human Language Technologies

Session 13: Neural networks part 1, project peer group feedback

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October 6, 2026

Quiz

- Go to **Quizzes > Quiz 10-06** on Canvas.
 - Session 11 Multinomial logistic regression, text classification evaluation: J+M 4.7-4.10, 4.12
 - Session 12 Vector semantics, word2vec: J+M 5-5.2, 5.5-5.8, 5.10
- You have until **1:10pm** to complete it
- Allowed resources
 - Textbook
 - Your notes (on a computer or on paper)
 - Course slides and website
- Resources not allowed
 - Generative AI
 - Internet searches

Assessments

- Project deliverable 1 is due **this Thu Oct 8**
- Homework 2 on text classification is due **next Thu Oct 15**

Midterm OMET feedback: what helps learning

8 responses out of 51 students (15.7%)

- Textbook
- Quizzes (force to read book)
- Talk-to-neighbor exercises
- Top Hat questions

Midterm OMET feedback: what could be improved

- Math is not making sense, going too fast
 - More examples needed
 - Live problem solving of questions
- Slides too fast
- In-class Jupyter notebooks
 - Messy, don't work, or examples can feel rushed
 - What is the point behind them?
- More Top Hat questions (for examples, not just comprehension checks)
- Study group (is coming!)

Learning objectives: neural networks part 1, project peer group feedback

- Explain the **fundamental parts of neural networks**
- Describe **non-linear activation functions**
- Explain how **feedforward neural networks can act as classifiers**
- Project peer group feedback

Why neural networks?

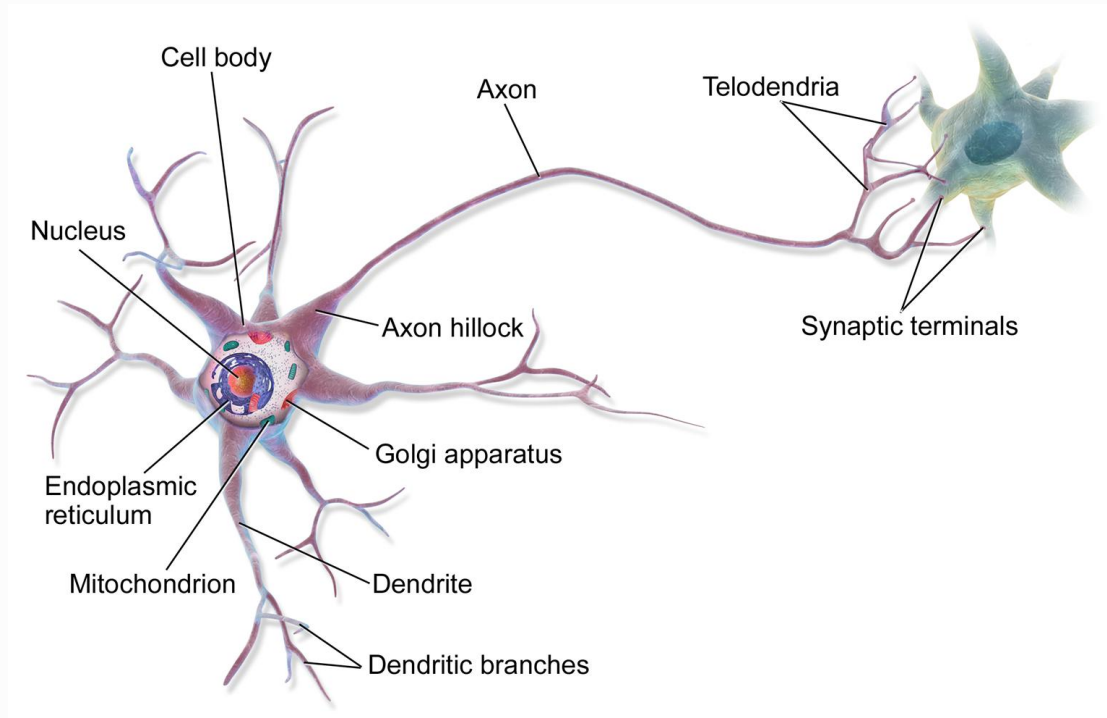
- Powerful classifiers for supervised learning
- At the heart of state-of-the-art machine learning systems in computer vision and NLP
- Large language models, search engines, speech recognition, ad and content recommendation systems



Pittsburgh Business Times

Neural network fundamentals

This is in your brain



By BruceBlaus - Own work, CC BY 3.0,
<https://commons.wikimedia.org/w/index.php?curid=28761830>

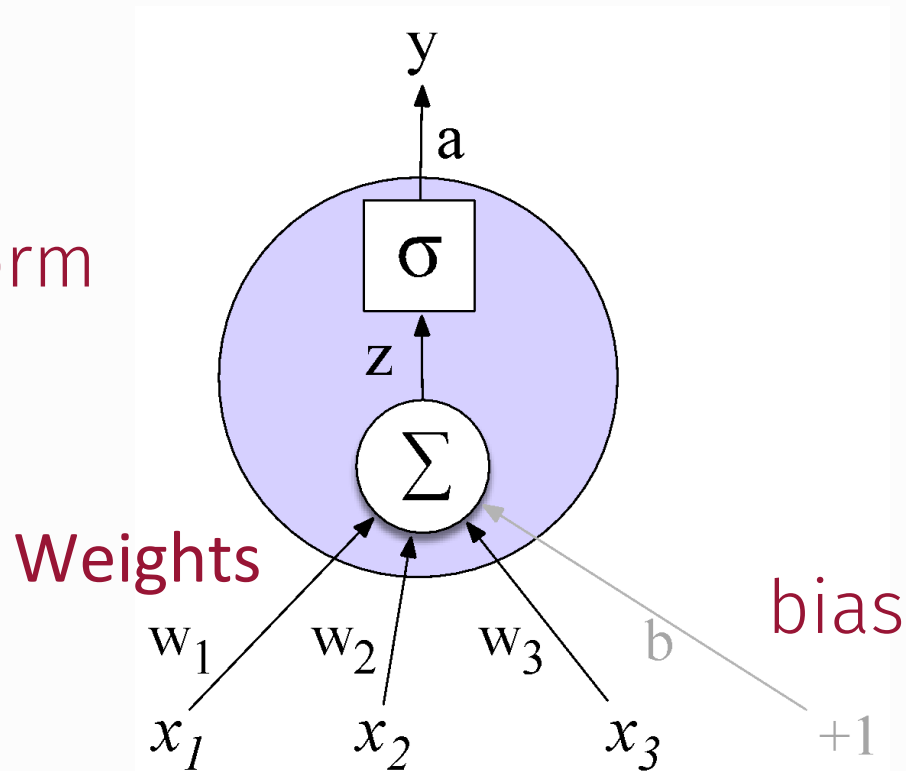
Neural network unit: This is not in your brain

Output value

Non-linear transform

Weighted sum

Input layer



The Variables in Our Very Important Formula

- x** A vector of features of n dimensions (like number of positive sentiment words, length of document, etc.)
- w** A vector of weights of n dimensions specifying how discriminative each feature is
- b** A scalar bias term that shifts z
- z** The raw score
- y** A random variable (e.g., $y = 1$ means positive sentiment and $y = 0$ means negative sentiment)

The Fundamentals

The fundamental equation that describes a unit of a neural network should look very familiar:

$$z = b + \sum_i w_i x_i \quad (1)$$

Which we will represent as

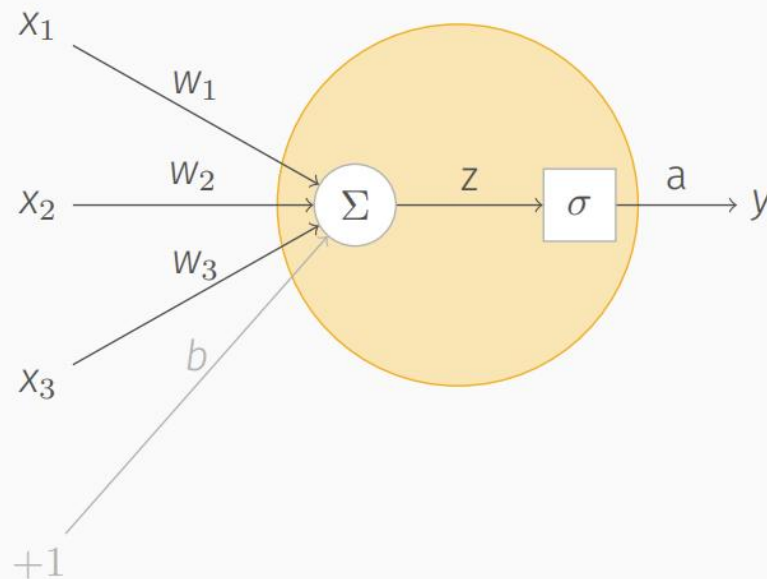
$$z = \mathbf{w} \cdot \mathbf{x} + b \quad (2)$$

But we do not use z directly. Instead, we pass it through a non-linear function, like the sigmoid function:

$$y = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (3)$$

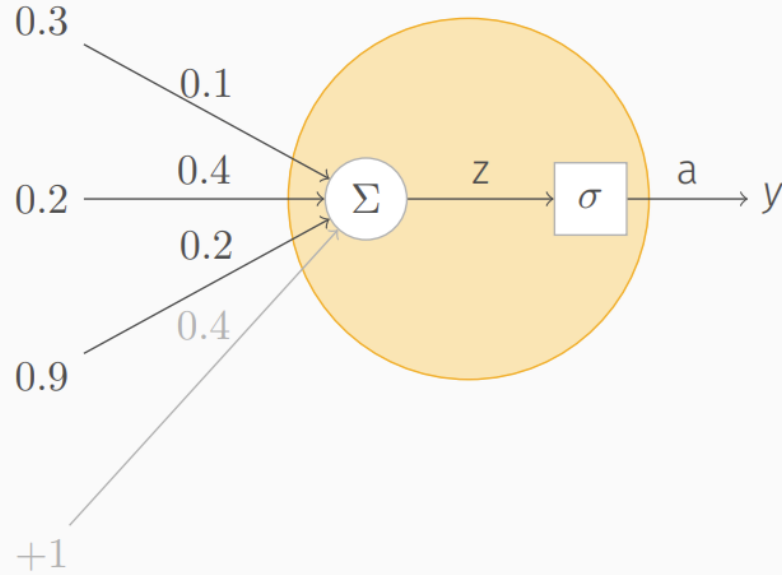
(which has some nice properties even though, in practice, we will prefer other functions like tanh and ReLU).

A Unit Illustrated

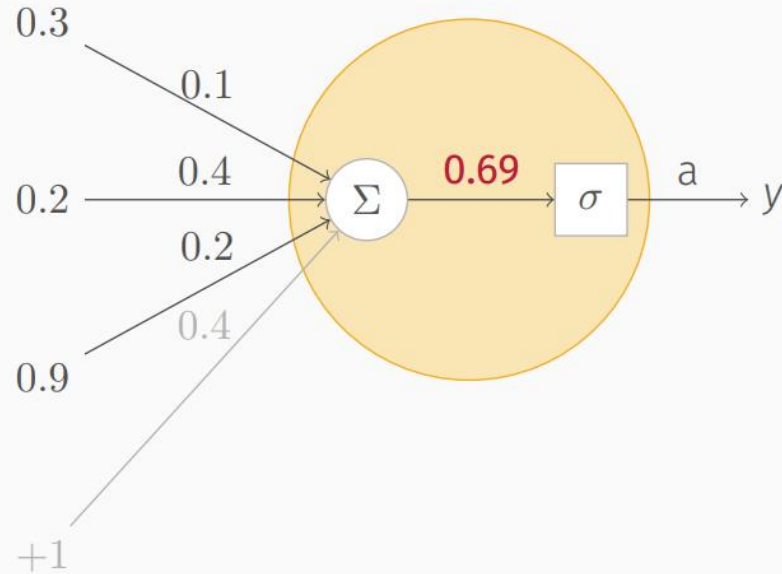


Take, for example, a scenario in which our unit has the weights $[0.1, 0.4, 0.2]$ and the bias term 0.4 and the input vector x has the values $[0.3, 0.2, 0.9]$.

Filling in the Input Values and Weights

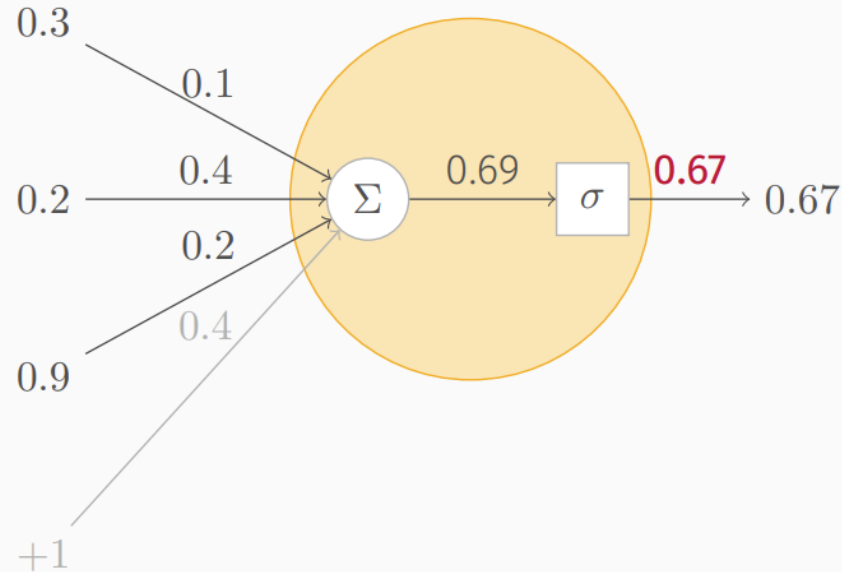


Multiplying the Input Values and Weights and Summing Them (with the Bias Term)



$$z = x_1w_1 + x_2w_2 + x_3w_3 + b = 0.1(0.3) + 0.4(0.2) + 0.2(0.9) + 0.4 = 0.69 \quad (4)$$

Applying the Activation Function (Sigmoid)



$$y = \sigma(0.69) = \frac{1}{1 + e^{-0.69}} = 0.67 \quad (5)$$

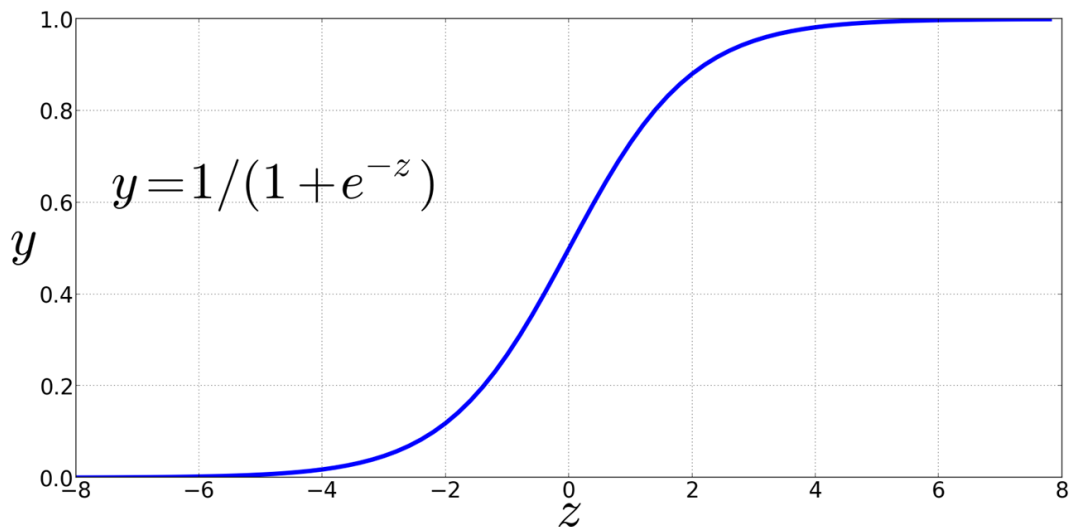
Non-linear activation functions

Non-Linear Activation Functions

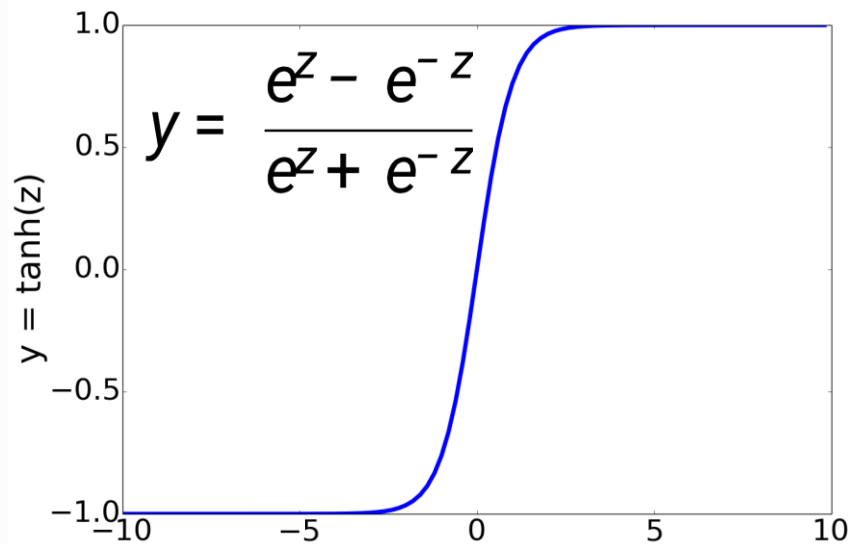
We're already seen the sigmoid for logistic regression:

Sigmoid

$$y = \sigma(z) = \frac{1}{1 + e^{-z}}$$

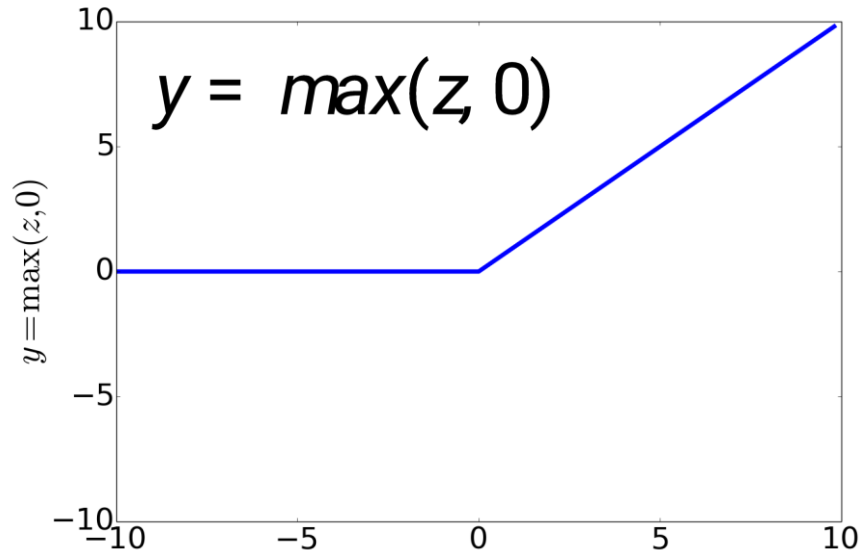


Nonlinear activation functions besides sigmoid



tanh

Most common:



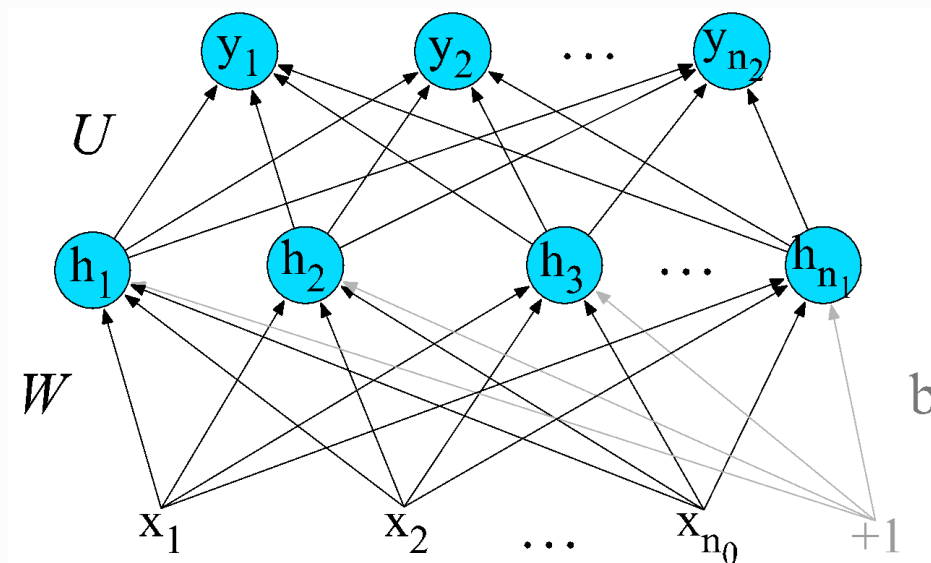
ReLU
Rectified Linear

Feedforward neural networks

Adding multiple units to a neural network increases its power to learn patterns in data. **Feedforward Neural Nets (FFNNs or MLPs)**

Feedforward Neural Networks

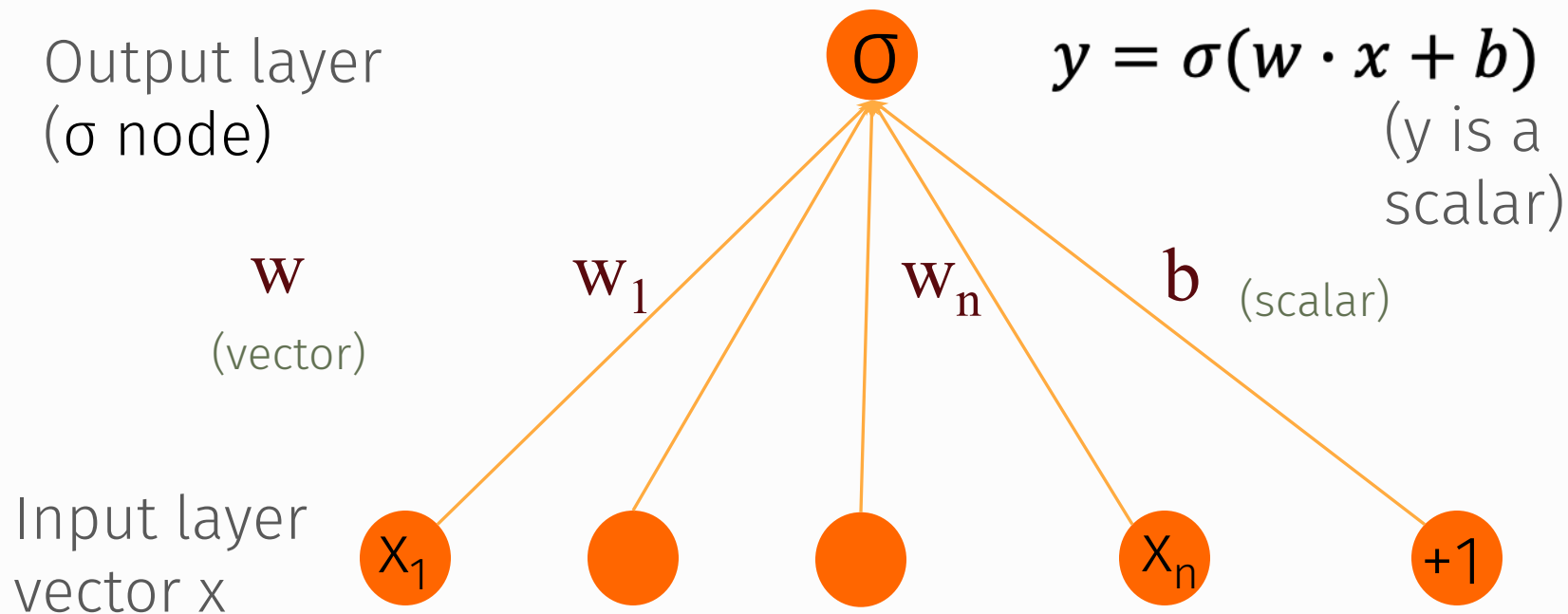
Can also be called **multi-layer perceptrons** (or **MLPs**) for historical reasons



The simplest FFNN is just binary logistic regression
(INPUT LAYER = feature vector)

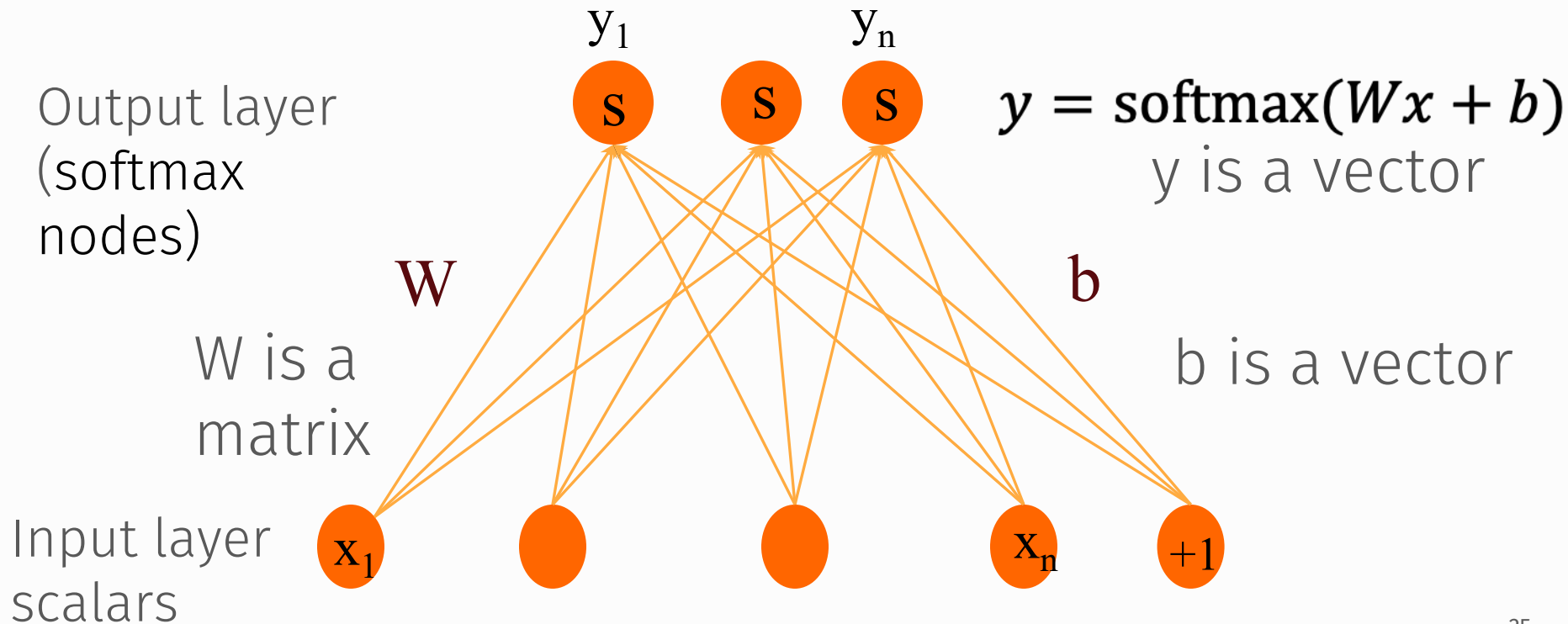
Binary Logistic Regression as a 1-layer Network

(we don't count the input layer in counting layers!)



Multinomial Logistic Regression as a 1-layer Network

Fully connected single layer network



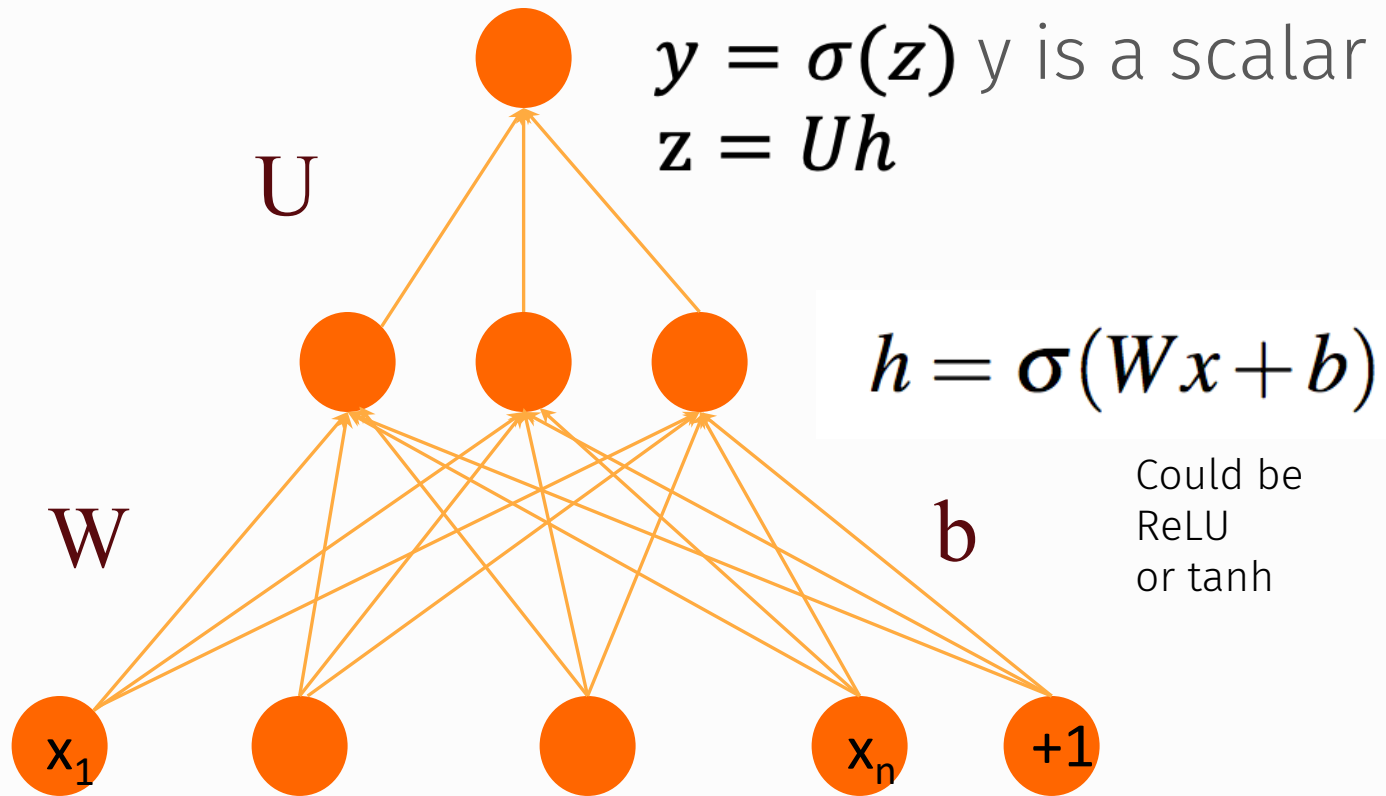
The real power comes when multiple layers are added

Two-Layer Network with scalar output

Output layer
(σ node)

hidden units
(σ node)

Input layer
(vector)

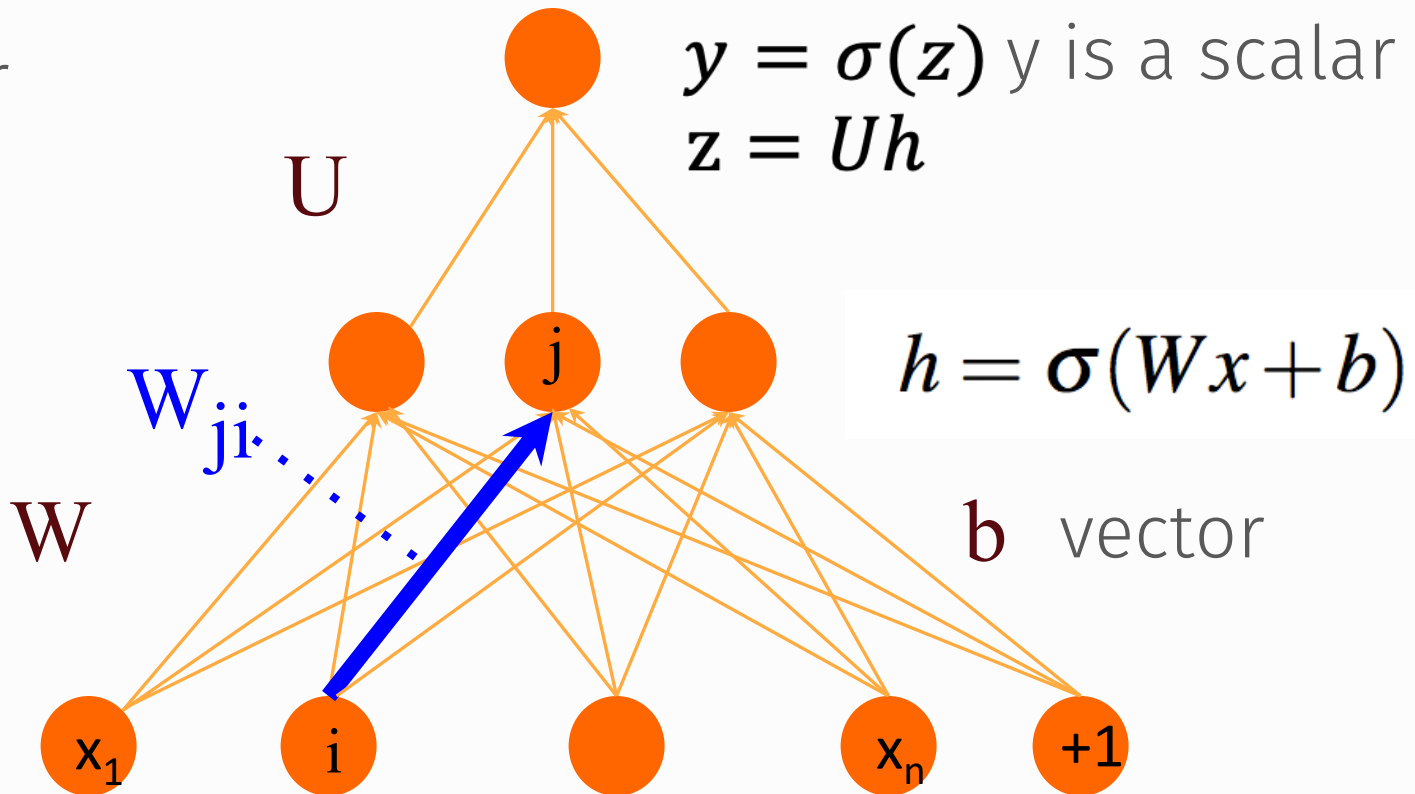


Two-Layer Network with scalar output

Output layer
(σ node)

hidden units
(σ node)

Input layer
(vector)

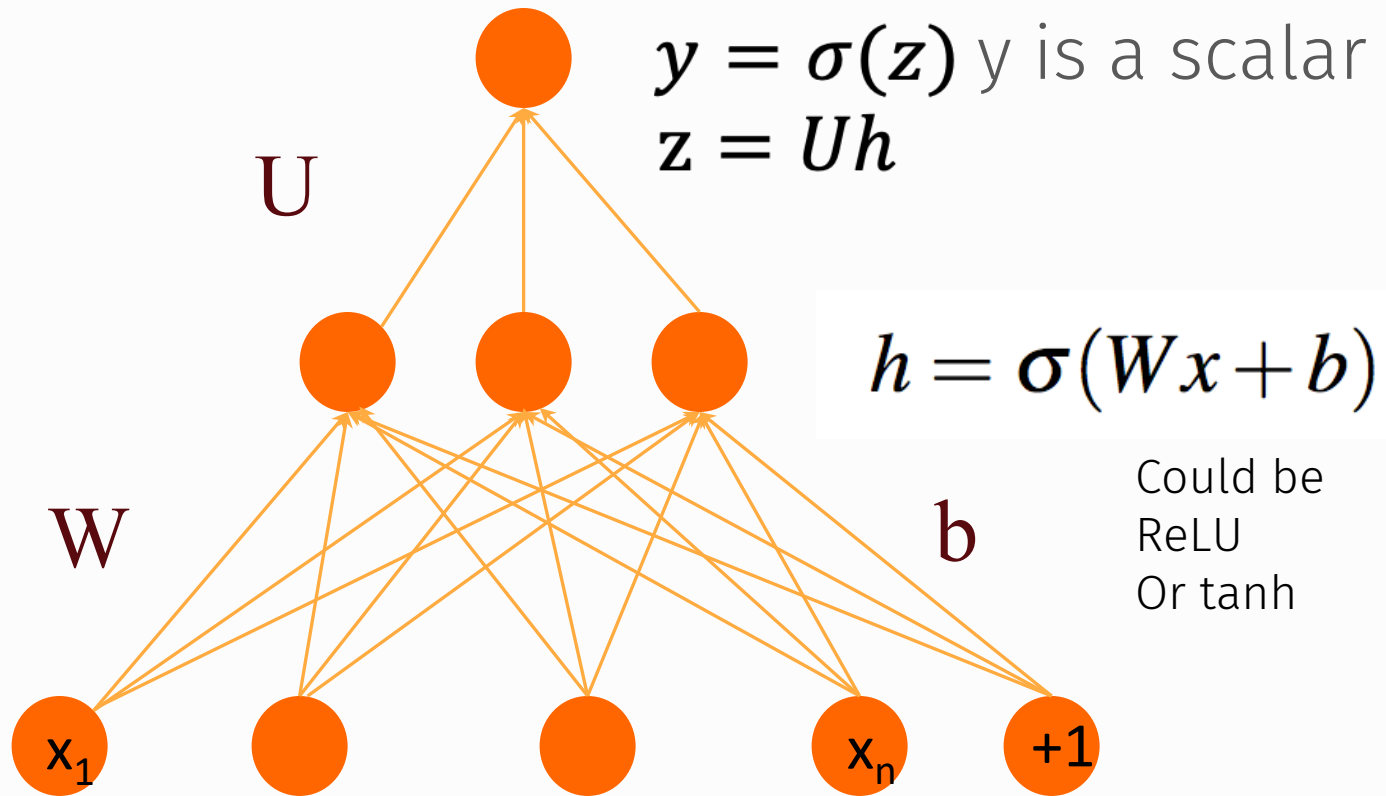


Two-Layer Network with scalar output

Output layer
(σ node)

hidden units
(σ node)

Input layer
(vector)

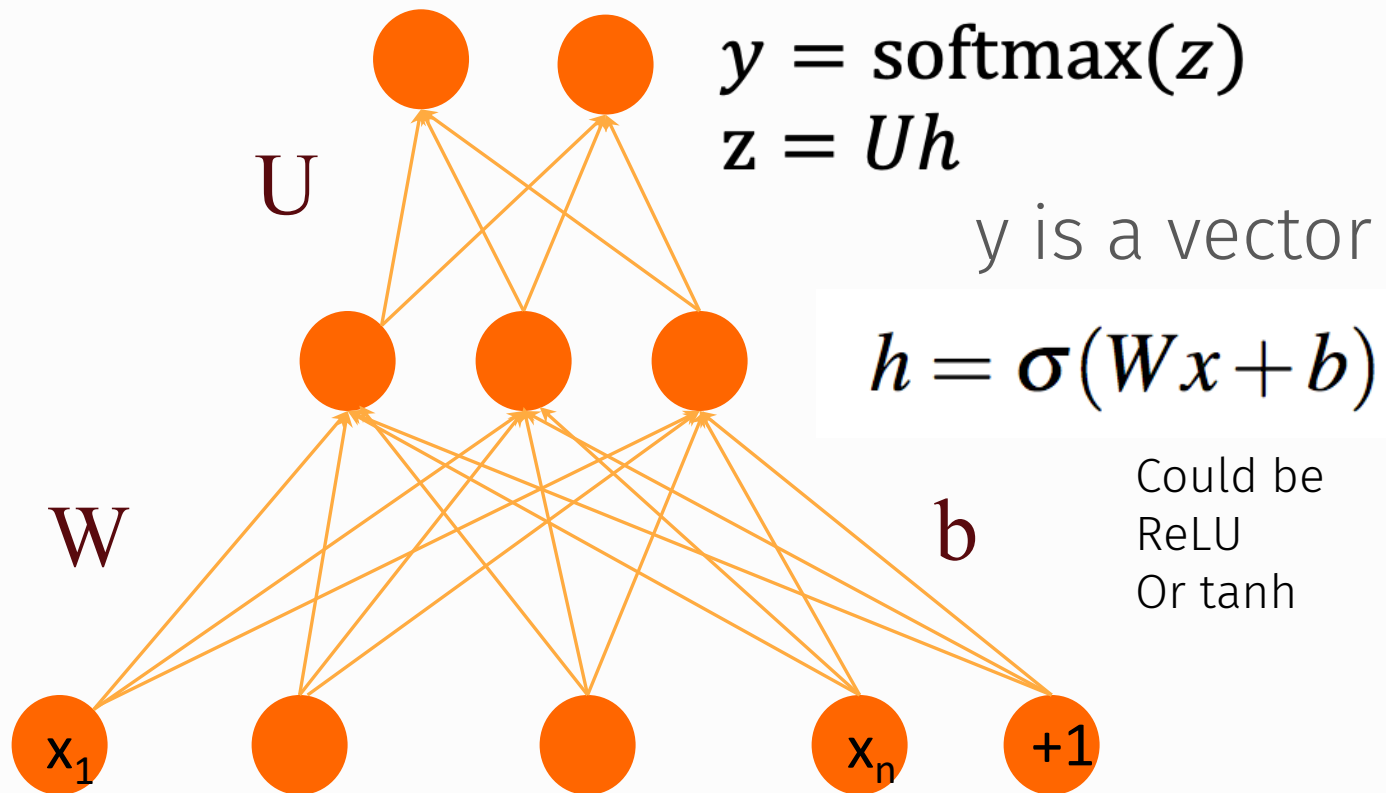


Two-Layer Network with softmax output

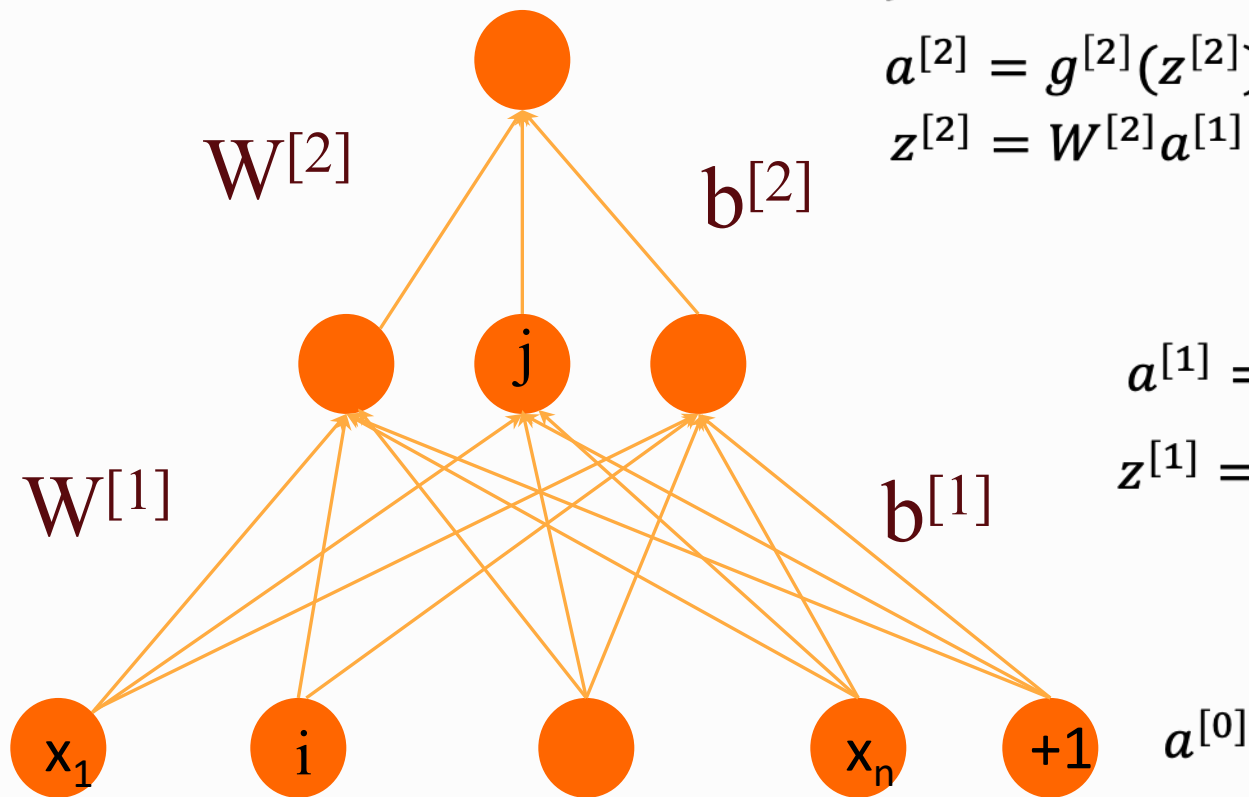
Output layer
(σ node)

hidden units
(σ node)

Input layer
(vector)



Multi-layer Notation



$$y = a^{[2]}$$

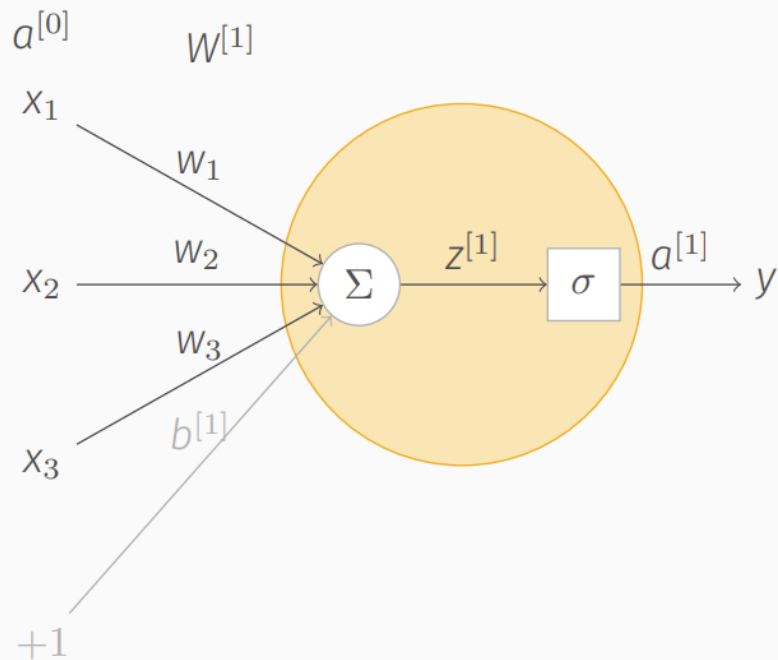
$$a^{[2]} = g^{[2]}(z^{[2]}) \quad \text{sigmoid or softmax}$$

$$z^{[2]} = W^{[2]}a^{[1]} + b^{[2]}$$

$$a^{[1]} = g^{[1]}(z^{[1]}) \quad \text{ReLU}$$

$$z^{[1]} = W^{[1]}a^{[0]} + b^{[1]}$$

A Forward Pass in Terms of Multi-Layer Notation



```
for each  $i \in 1..n$  do  
     $z^{[i]} \leftarrow W^{[i]}a^{[i-1]} + b^{[i]}$   
     $a^{[i]} \leftarrow g^{[i]}(z^{[i]})$   
end for  
 $\hat{y} \leftarrow a^{[n]}$ 
```

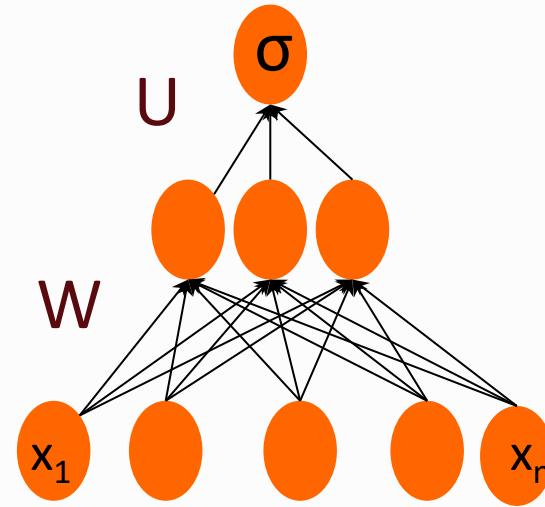
Feedforward neural nets as classifiers

Classification: Sentiment Analysis

We could do exactly what we did with logistic regression

Input layer are binary features as before

Output layer is 0 or 1

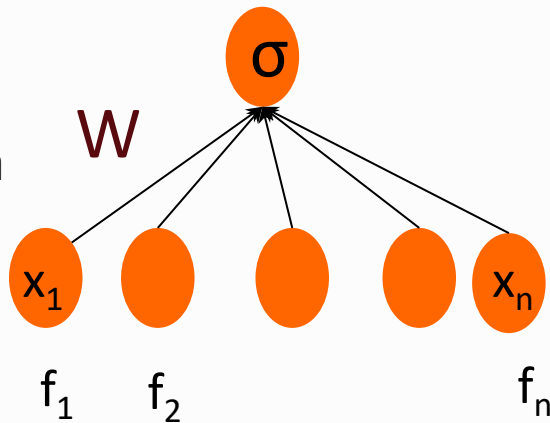


Sentiment Features

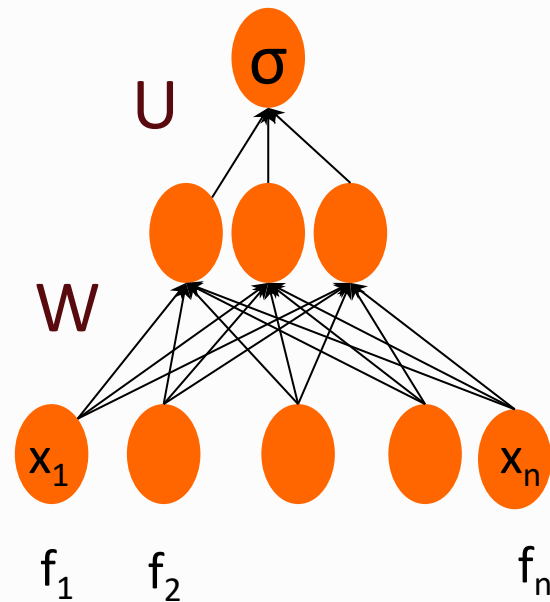
Var	Definition
x_1	$\text{count}(\text{positive lexicon}) \in \text{doc}$
x_2	$\text{count}(\text{negative lexicon}) \in \text{doc}$
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$
x_4	$\text{count}(\text{1st and 2nd pronouns}) \in \text{doc}$
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$
x_6	$\log(\text{word count of doc})$

Feedforward nets for simple classification

Logistic Regression



2-layer feedforward network



Just adding a hidden layer to logistic regression

- allows the network to use non-linear interactions between features
- which may (or may not) improve performance.

Conclusion: neural networks part 1

- Neural networks are machine learning systems with “layers” of weighted sums and nonlinear functions
 - Take a vector representing a datapoint as input (feature vector)
 - Produce probabilities over labels (from a softmax) as output
- “Activation functions” are nonlinear functions that enable neural networks to model complex, nonlinear relationships between input and output
 - ReLU, sigmoid, tanh

Project peer group feedback

Project peer group feedback

1. Find another group to work with. Decide which project group will go first as the presenting group
2. Present an overview of your project: 5 min
3. Other group asks clarifying questions
4. Presenting group asks for any advice or guidance from the other group on lingering questions about the project proposal
5. Switch groups (the other group becomes the presenting group) when Michael says